Total No. of Questions : 8]

SEAT No. :

PC2376

[Total No. of Pages : 2

[6354]-493
B.E. (Computer Engineering)
INFORMATION RETRIEVAL
(2019 Pattern) (Semester - VII) (Elective - IV) (410245 A)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) List out the General-Purpose Data Compression techniques and explain with suitable example. **[6]**

b) Describe in detail Invalidation List, Garbage Collection, Document Modifications in document deletion. **[6]**

c) Write a short note on. **[6]**

i) Modeling and coding

ii) Arithmetic Coding

OR

Q2) a) Explain software Architecture of the IR system in detail. **[6]**

b) Explain in details: **[6]**

i) Decoding performance

ii) Document Reordering

c) Describe data compression with Huffman coding. **[6]**

Q3) a) Explain categorization and filtering with any two detailed examples. **[7]**

b) Describe passage retrieval and ranking with example. **[5]**

c) Explain the Information-Theoretic Model in detail. **[5]**

OR

Q4) a) Explain probabilistic Classifiers & Generalized Linear Models. **[7]**

b) Explain Relevance Feedback Technique with suitable diagram. **[5]**

c) Describe Language Models and Smoothing. **[5]**

Q5) a) Explain Measuring effectiveness like Traditional effectiveness measure and the text retrieval conference (TREC) with suitable examples. **[6]**

b) Write a short note on: **[6]**

i) Nontraditional effectiveness measures.

ii) Measuring efficiency

c) Explain query scheduling with suitable examples. **[6]**

OR

Q6) a) Write a short note on: **[6]**

i) Minimizing adjudication effort.

ii) Using statistics in evaluation

b) Explain caching with suitable examples. **[6]**

c) Differentiate between Redis and Memcached. **[6]**

Q7) a) Describe Map reduce with suitable examples. **[6]**

b) Write a short note on: **[6]**

i) The structure of the web

ii) Python Scrapy

c) Describe web crawler with its components. **[5]**

OR

Q8) a) Describe Parallel Query Processing with suitable examples. **[6]**

b) Explain the following term:

i) Static ranking

ii) Dynamic ranking **[6]**

c) Write a short note on Evaluation web search. **[5]**

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P-6559

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[6181]-109

Fourth Year B.E. (Computer Engineering)

INFORMATION RETRIEVAL

(2019 Pattern) (Semester - VII) (410245 A) (Elective - IV)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) ***Solve Q.1 or Q.2 and Q.3 or Q.4 and Q.5 or Q.6 and Q.7 or Q.8.***
2) ***Figures to the right indicate full marks.***
3) ***Neat diagram must be drawn whenever necessary.***
4) ***Assume suitable data if necessary.***

Q1) a) What is Data Compression? Explain Modeling and Coding. **[9]**

b) Write a short note on : **[9]**

i) Invalidation List

ii) Garbage Collection

iii) Document Modifications

OR

Q2) a) Write a short note on : **[9]**

i) Decoding Performance

ii) Document Reordering

iii) Arithmetic Coding

b) Differentiate between Contiguous Inverted Lists & Noncontiguous Inverted. **[5]**

c) Differentiate between Non parametric Gap Compression & Parametric Gap Compression. **[4]**

Q3) a) Explain Categorization and Filtering with any two detailed Examples of each. **[8]**

b) Write a short note on Generalized Linear Models. **[9]**

OR

Q4) a) Explain Probabilistic Classifiers, Information-Theoretic Model in detail. **[9]**

b) Describe Language Models and Smoothing. **[8]**

Q5) a) Explain Traditional effectiveness measure and The Text Retrieval Conference (TREC) with suitable examples. **[9]**

b) Write a short note on : **[9]**

i) Nontraditional effectiveness measures

ii) Measuring efficiency

OR

Q6) a) What is Scheduling and Caching in Measuring Efficiency?. Explain in detail. **[9]**

b) Write a short note on : **[9]**

i) Using statistics in evaluation

ii) Minimizing adjudication Effort

Q7) a) Explain in detail Parallel Query Processing with suitable examples. **[9]**

b) Write a short note on : **[8]**

i) Static ranking

ii) Dynamic ranking

OR

Q8) a) Describe MapReduce with suitable examples. **[9]**

b) Write a short note on : **[8]**

i) Evaluation web search

ii) Web Crawlers

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Total No. of Questions : 8]

SEAT No. :

P553

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[6004]-488

B.E. (Computer Engineering)

INFORMATION RETRIEVAL

(2019 Pattern) (Semester - VII) (Elective - IV) (410245 (A))

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Figures to the right side indicate full marks.*
3) *Neat diagram must be drawn wherever necessary.*
4) *Assume suitable data if necessary.*

Q1) a) Enlist the General-Purpose Data Compression techniques and explain with suitable examples. **[9]**

b) Write a short note on. **[9]**

i) Nonparametric Gap Compression.

ii) Parametric Gap Compression.

iii) Context-Aware Compression Methods.

OR

Q2) a) Write a short note on. **[9]**

i) Modeling and Coding.

ii) Huffman Coding.

iii) Arithmetic Coding.

b) Describe Dynamic Inverted Indices like Incremental Index Updates, Contiguous Inverted Lists and Noncontiguous Inverted. **[9]**

Q3) a) Explain Categorization and Filtering with any two detailed Examples. **[9]**

b) Explain the Information-Theoretic Model in detail. **[8]**

OR

Q4) a) Explain probabilistic Classifiers & Generalized Linear Models. **[9]**

b) Describe Language Models and Smoothing. **[8]**

Q5) a) Explain MeasuringEffectiveness like Traditional effectiveness measure and the text retrieval conference (TREC) with suitable examples. **[9]**

b) Write a short note on: **[9]**

i) Nontraditional effectiveness measures.

ii) Measuring efficiency.

OR

Q6) a) Explain Query Scheduling and Caching with suitable examples. **[9]**

b) Write a short note on: **[9]**

i) Using statistics in evaluation.

ii) Minimizing adjudication Effort.

Q7) a) Describe Parallel Query Processing with suitable examples. **[8]**

b) Write a short note on: **[9]**

i) The structure of the web.

ii) Quires and Users.

iii) Static ranking.

OR

Q8) a) Describe Map Reduce with suitable examples. **[9]**

b) Write a short note on: **[8]**

i) Evaluation web search.

ii) Web Crawlers.

iii) Web crawler libraries.

iv) Dynamic ranking.

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Total No. of Questions : 8]

SEAT No. :

PA-918

[Total No. of Pages : 2

[5927]-350

B.E. (Computer Engineering)

INFORMATION RETRIEVAL

(2019 Pattern) (Semester - VII) (410245 A) (Elective - IV)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates :

1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Figures to the right indicate full marks.*
3) *Neat diagrams must be drawn wherever necessary.*
4) *Assume sitable data, if necessary.*

Q1) a) Explain Data Compression with Huffman Coding. **[9]**

b) What are the Context-Aware Compression Methods? Explain it in detail. **[9]**

OR

Q2) a) Write a short note on : **[9]**

i) Decoding Performance

ii) Document Reordering

iii) Arithmetic Coding

b) Explain in detail Invalidation List, Garbage Collection, Document Modifications in document deletion. **[9]**

Q3) a) Explain Categorization and Filtering with any two detailed Examples. **[9]**

b) Explain the Information-Theoretic Model in detail. **[8]**

OR

Q4) a) Explain Probabilistic Classifiers & Generalized Linear Models. **[9]**

b) Describe Language Models and Smoothing. **[8]**

Q5) a) Explain Traditional effectiveness measure and The Text Retrieval Conference (TREC) with suitable examples. **[9]**

b) Write a short note on : **[9]**

i) Nontraditional effectiveness measures

ii) Measuring efficiency

OR

Q6) a) What is Scheduling and Caching in Measuring Efficiency?. Explain in detail. **[9]**

b) Write a short note on : **[9]**

i) Using statistics in evaluation

ii) Minimizing adjudication Effort

Q7) a) Describe Parallel Query Processing with suitable examples. **[8]**

b) Write a short note on : **[9]**

i) The structure of the web

ii) Quires and Users

iii) Static ranking

OR

Q8) a) Describe MapReduce with suitable examples. **[9]**

b) Write a short note on : **[8]**

i) Evaluation web search

ii) Web Crawlers

iii) Web crawler libraries

iv) Dynamic ranking

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